

GAYA, Niger

WMO: 610990

Lat: 11.883N

Lon: 3.450E

Elev: 663

StdP: 14.35

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 65.8	(c) 67.7	(d) 18.8	(e) 14.5	(f) 82.5	(g) 21.4	(h) 16.5	(i) 82.6	(j) 20.6	(k) 73.6	(l) 18.7	(m) 74.1	(n) 7.0	(o) 40	(p) 0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 21.1	(c) 106.5	(d) 71.4	(e) 105.1	(f) 71.7	(g) 103.5	(h) 72.0	(i) 81.6	(j) 92.8	(k) 80.9	(l) 92.2	(m) 80.3	(n) 91.6	(o) 6.1	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	154.2	86.6	78.2	150.2	86.0	77.6	147.1	85.5	45.9	93.5	45.1	92.6	44.5	91.6	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.5	(b) 14.1	(c) 12.1	(e) 60.3	(f) 109.6	(g) 4.1	(h) 1.6	(i) 57.3	(j) 110.8	(k) 54.9	(l) 111.7	(m) 52.6	(n) 112.6	(o) 49.7	(p) 113.8		
			DB													
			WB													
				60.3	109.6	4.1	1.6	57.3	110.8	54.9	111.7	52.6	112.6	49.7	113.8	
				45.5	83.9	2.4	1.2	43.7	84.8	42.3	85.5	40.9	86.1	39.2	87.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	85.4	80.3	85.5	91.7	94.0	90.7	86.0	82.6	80.6	82.2	85.7	85.1	81.1
	DBStd	5.54	4.03	4.77	3.44	3.10	4.12	3.79	3.17	2.64	2.85	2.37	2.30	3.37
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12937	939	995	1293	1321	1261	1079	1009	949	967	1107	1054	963
	CDD65	7462	474	575	828	871	796	629	544	484	517	642	604	498
	CDH74	98645	5188	7741	13047	14140	12080	8411	6090	4604	5615	8164	7991	5574
	CDH80	56276	2619	4540	8719	9853	7769	4538	2572	1547	2322	4309	4628	2860
(14) Wind	WSAvg	5.1	7.7	7.1	6.5	5.3	5.5	4.5	3.7	3.6	3.3	3.3	4.2	6.8
(15) Precipitation	PrecAvg	31.1	0.0	0.1	0.1	0.6	2.6	4.4	7.1	8.9	6.4	0.7	0.0	0.0
	PrecMax	40.9	0.0	1.7	1.3	2.3	8.4	8.3	13.8	13.3	9.9	2.3	0.1	0.2
	PrecMin	18.6	0.0	0.0	0.0	0.0	0.2	1.7	2.3	5.0	1.3	0.0	0.0	0.0
	PrecStd	4.4	0.0	0.3	0.3	0.7	1.6	1.8	2.6	2.6	2.0	0.7	0.0	0.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.8	105.1	107.6	108.8	106.7	101.5	96.9	92.6	95.2	100.2	101.6	99.6
		MCWB	65.5	68.1	69.2	72.4	76.0	78.1	78.9	79.6	80.6	73.5	70.0	66.8
	2%	DB	97.0	102.6	105.8	107.1	104.3	99.0	93.6	90.5	93.2	98.2	99.9	97.5
		MCWB	63.8	67.5	68.7	71.9	77.1	78.3	78.6	79.2	80.1	74.9	69.8	65.7
	5%	DB	94.7	100.1	104.4	105.8	102.3	97.0	91.8	88.9	91.6	96.6	98.5	95.4
		MCWB	62.7	66.1	68.5	72.5	77.7	78.3	78.6	78.8	79.6	76.0	69.0	64.8
	10%	DB	92.1	97.6	102.8	104.1	100.3	94.8	90.0	87.3	89.8	94.8	96.9	92.9
		MCWB	61.8	64.9	68.0	73.6	77.9	78.4	78.4	78.4	79.1	77.2	68.1	63.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	76.6	78.6	80.8	81.9	81.9	81.3	81.0	82.1	82.4	77.8	70.0
		MCDB	89.8	94.3	98.4	99.5	96.7	93.5	91.1	89.4	91.5	93.1	92.2	94.6
	2%	WB	66.7	73.0	76.5	79.7	80.9	80.9	80.3	80.1	81.2	81.4	76.0	67.9
		MCDB	92.8	90.4	96.3	98.3	96.1	92.6	89.6	87.9	90.2	92.1	90.4	93.4
	5%	WB	64.1	69.2	75.1	78.8	80.1	80.0	79.7	79.4	80.5	80.5	74.4	66.4
		MCDB	92.5	96.2	94.7	97.0	95.2	91.7	88.7	86.8	89.3	91.2	88.8	92.0
	10%	WB	62.2	66.4	73.2	77.9	79.5	79.3	78.9	78.8	79.8	79.6	72.7	64.7
		MCDB	90.9	96.6	94.3	95.8	94.3	90.7	87.8	85.8	88.0	89.9	87.7	90.5
(35) Mean Daily Temperature Range	5% DB	MDBR	23.7	23.3	22.7	21.1	18.7	16.7	14.4	12.7	14.8	19.7	25.9	24.3
		MCDBR	26.5	25.4	24.2	22.6	20.7	19.0	16.7	15.1	17.3	22.3	27.1	27.1
		MCWBR	11.3	10.9	10.6	7.9	5.5	5.1	5.1	5.4	6.8	6.2	8.1	10.7
	5% WB	MCDBR	25.5	23.0	21.2	20.1	18.3	17.0	15.4	14.1	16.3	18.5	24.1	26.2
		MCWBR	12.2	11.0	10.6	6.2	5.7	5.8	5.7	5.9	6.9	6.6	8.0	11.0
(40) Clear-Sky Solar Irradiance	taub		0.637	0.710	0.776	0.745	0.679	0.628	0.568	0.550	0.557	0.621	0.569	0.552
	taud		1.673	1.518	1.426	1.555	1.666	1.784	1.970	2.009	1.973	1.790	1.805	1.823
	Ebn at Noon		217	208	198	204	214	222	237	245	245	226	233	235
	Edh at Noon		77	93	104	91	80	70	58	57	59	70	67	65
(44) All-Sky Solar Radiation	RadAvg		1727	1820	1905	1963	1905	1814	1702	1584	1758	1885	1821	1724
	RadStd		83	80	97	69	57	73	66	76	71	61	46	80

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page